ROCSCIENCE

Case2_Search_Iter_3
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:45:25 am
Software Version: 9.041

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Slide2 Analysis Information

Case2_Search_Iter_3

Project Summary

File Name:	Case2_Search_Iter_3.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.543s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:45:25 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	7
Tolerance:	0.001
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	3
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	20 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	2.106250
Center:	14.468, 24.384
Radius:	15.140
Left Slip Surface Endpoint:	9.744, 10.000
Right Slip Surface Endpoint:	27.952, 17.500
Resisting Moment:	12793.7 kN-m
Driving Moment:	6074.15 kN-m
Total Slice Area:	57.9061 m2
Surface Horizontal Width:	18.2081 m
Surface Average Height:	3.18023 m

Method: spencer

FS	2.100420
Center:	14.468, 24.384
Radius:	15.140
Left Slip Surface Endpoint:	9.744, 10.000
Right Slip Surface Endpoint:	27.952, 17.500
Resisting Moment:	12758.3 kN-m
Driving Moment:	6074.15 kN-m
Resisting Horizontal Force:	737.549 kN
Driving Horizontal Force:	351.143 kN
Total Slice Area:	57.9061 m2
Surface Horizontal Width:	18.2081 m
Surface Average Height:	3.18023 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	3675
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	3597
Number of Invalid Surfaces:	78

Error Codes
Error Code -108 reported for 55 surfaces
Error Code -111 reported for 23 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.10625

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.60116	43.2737	-13.1204	Material 1	20	20	12.8894	27.1483	19.6397	0	19.6397	16.6354	16.6354
2	2.60116	130.071	-3.1245	Material 1	20	20	18.3093	38.5639	51.0039	0	51.0039	50.0045	50.0045
3	2.60116	193.388	6.77368	Material 1	20	20	21.8938	46.1138	71.7472	0	71.7472	74.3477	74.3477
4	2.60116	232.472	16.8872	Material 1	20	20	23.6968	49.9113	82.1809	0	82.1809	89.3747	89.3747
5	2.60116	244.219	27.6007	Material 1	20	20	23.5895	49.6853	81.5596	0	81.5596	93.8922	93.8922
6	2.60116	219.994	39.5611	Material 1	20	20	21.0993	44.4405	67.1497	0	67.1497	84.5804	84.5804
7	2.60116	94.7051	54.4567	Material 1	20	20	12.7131	26.777	18.6196	0	18.6196	36.4142	36.4142

Global Minimum Query (spencer) - Safety Factor: 2.10042

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.60116	43.2737	-13.1204	Material 1	20	20	14.1642	29.7507	26.7898	0	26.7898	23.4884	23.4884
2	2.60116	130.071	-3.1245	Material 1	20	20	19.7068	41.3925	58.7754	0	58.7754	57.6997	57.6997
3	2.60116	193.388	6.77368	Material 1	20	20	22.7328	47.7485	76.2386	0	76.2386	78.9388	78.9388
4	2.60116	232.472	16.8872	Material 1	20	20	23.6956	49.7708	81.7946	0	81.7946	88.9881	88.9881
5	2.60116	244.219	27.6007	Material 1	20	20	22.7342	47.7514	76.2465	0	76.2465	88.132	88.132
6	2.60116	219.994	39.5611	Material 1	20	20	19.6854	41.3476	58.6523	0	58.6523	74.9149	74.9149
7	2.60116	94.7051	54.4567	Material 1	20	20	12.2758	25.7844	15.8924	0	15.8924	33.075	33.075

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.10625

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	9.74419	10	0	0	0
2	12.3453	9.39371	45.4242	0	0
3	14.9465	9.25172	100.277	0	0
4	17.5477	9.56068	135.042	0	0
5	20.1488	10.3503	131.767	0	0
6	22.75	11.7102	82.196	0	0
7	25.3512	13.8591	-7.23579	0	0
8	27.9523	17.5	0	0	0

Global Minimum Query (spencer) - Safety Factor: 2.10042

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	9.74419	10	0	0	0
2	12.3453	9.39371	53.0826	17.8295	18.5663
3	14.9465	9.25172	112.685	37.8487	18.5662
4	17.5477	9.56068	148.257	49.7969	18.5663
5	20.1488	10.3503	145.299	48.8032	18.5662
6	22.75	11.7102	100.742	33.8376	18.5664
7	25.3512	13.8591	25.9061	8.70141	18.5663
8	27.9523	17.5	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 0 35, 0 35, 17.5 25, 17.5 10, 10 0, 10